

University of Bologna

Curriculum - Data Science

Does climate change exacerbate the
existing economic and social
differences between northern and
southern Italy?

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Population and Climate Change

A.Y. 2024-2025

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Introduction

Socioeconomic differences between southern and northern Italy

The North and South division has been a socioeconomic feature of Italy's since its unification in 1861. The gap in GDP per capita between the South and the Central-North has been fixed around 40-45% for decades, reflecting disparities that are structural and institutional [11].

In 2023, these disparities remain apparent in almost all economic and social indicators. This imbalance is also mirrored in per capita income, with residents in the North earning an average of €24,000 per year in available income, compared to just €16,000 in the South, evidencing the unequal distribution of wealth [3][1].

Employment statistics reveal additional divisions. Approximately 70% of working-age people in the North are employed, compared to only 50% in the South. Youth and long-term unemployment is particularly severe in southern regions such as Calabria and Sicily, where unemployment rates exceed 16%, well above the national average [8]. The structure of the economy is also significantly different. Northern Italy is home to most of the country's industrial base and is home to world-renowned brands. In contrast, the South remains heavily dependent on agriculture and public sector employment, which tend to offer lower wages and limited opportunities for economic advancement. This divergence is partly due to historical patterns of infrastructure development and market integration. Northern regions enjoy better transport networks and greater proximity to European markets, which have fostered industrial growth since at least the mid-20th century [6].

Living standards reflect these differences. Northern regions such as Lombardy have Human Development Index (HDI) scores comparable to those of developed countries such as Germany and the United Kingdom. Meanwhile, southern regions such as Calabria and Sicily have HDI levels similar to those of middle-income countries such as Argentina or Turkey. Poverty is also more widespread in the south. For example, in 2018, Campania had over 500 poor households per 1,000 residents, while richer regions such as Tuscany had fewer than 100 [10][14].

Primary research questions

Considering the context previously described, the research question of the project came up naturally. Does climate change exacerbate the existing economic and social differences between northern and southern Italy? Given the well-established disparities in terms of income, employment, industrial development and standard of living, climate change can act as a multiplying factor that has a disproportionate impact on the more vulnerable south. The economy of southern Italy, which is heavily dependent on agriculture and public sector employment, is likely to be more sensitive to temperature variations, water availability and extreme weather events than the industrialised North. Furthermore, weaker infrastructure and limited adaptive capacity in southern regions could hamper their ability to respond effectively to climate-related shocks, potentially exacerbating poverty, unemployment and migration. In contrast, northern Italy, with its diversified economy, stronger institutions and greater financial resources, may be better positioned to mitigate climate impacts and maintain economic resilience. This raises the crucial question of whether climate change risks widening the socioeconomic gap, reinforcing regional inequalities and posing a serious challenge to Italy's socioeconomic cohesion and long-term development. This project aims to explore these dynamics by analysing climate vulnerability alongside economic, demographic and social indicators in Italian regions [17][12].

Secondary research question

Since its adoption in 2015, the Paris Agreement has become the benchmark for international climate policy, with the goal of limiting global temperature rise to well below 2°C above pre-industrial levels, with an ambitious target of 1.5°C. The agreement also commits signatory countries to achieving net zero greenhouse gas emissions in the second half of the 21st century.

The agreement has had a measurable impact on global climate governance: projected warming by 2100 has decreased from pre-agreement estimates of 3.7-4.8°C to around 2.4-2.7°C, or even 1.7-2.1°C if all long-term commitments are met. It has stimulated widespread national commitments, including net-zero targets by major emitters such as the European Union, China and Japan, and has led to near-universal participation thanks to its flexible and non-binding structure [13].

Despite this progress, significant challenges remain. Current national commitments are insufficient to achieve the 1.5°C target, and global greenhouse gas emissions continue to rise, with 2024 marking the warmest year on record. Furthermore, the agreement is based on voluntary compliance, with no legally binding enforcement mechanisms, raising concerns about the credibility and sufficiency of national commitments.

Although the Paris Agreement has undoubtedly changed the trajectory of global cli-

mate policy and promoted greater international cooperation, it has not yet succeeded in achieving the reduction in emissions needed to avoid the most severe impacts of climate change [15].

Considering the significant impact that the Paris Agreement has had on international climate policy, another question of interest has arisen: Is there statistically significant evidence of a change in climate variable trends since the Paris Agreement was implemented in Italy?

Since the Paris Agreement entered into force internationally on 4 November 2016 and was formally implemented by Italy on 11 November 2016, any change in trend must have occurred after that date in order to be attributed to the Agreement.

Data used

In order to answer the research questions, a large amount of data was collected from various research institutions. Most of the data comes from Eurostat[4] (data on agriculture, demographics, economy, education and tourism), while the climatic data was obtained from Copernicus [2]. Lastly, some data on minor variables, which were nevertheless considered important for the purpose of the project, were obtained from Istat [9].

Given that the main objective of this study is to determine whether there is statistical evidence of a different impact of climate change among northern and southern Italy, all the data used in this report are at NUTS 2 level [5].

NUTS 2 data (Nomenclature of Territorial Units for Statistics - Level 2) represent a standardised classification of European intermediate regions, adopted by the European Union for the collection, analysis and dissemination of regional statistical data.

In Italy, the NUTS 2 level corresponds to the administrative regions. It includes the 20 Italian regions, with some exceptions based on organisational adjustments or European requirements, resulting in 21 different "districts".

All data are observed annually, with data from 2000 to 2023.

Exploratory graphs were created to understand how two of the most important variables, "GDP per capita" and "temperature", changed over time in the different Italian regions.

As can be seen in Figure 1 and in Figure 2, there has been a substantial change over time in both "GDP per capita" and "temperature".

This is certainly not sufficient to answer the research question posed. However, the results of this preliminary analysis are consistent with the vast literature existing on this topic, and indicate that a further and more in-depth investigation of the issue is necessary and could lead to interesting insights.

*Figure 1: GDP per capita per zone
(over time)*

*Figure 2: Average Temperature
per region (over time)*

Data manipulation

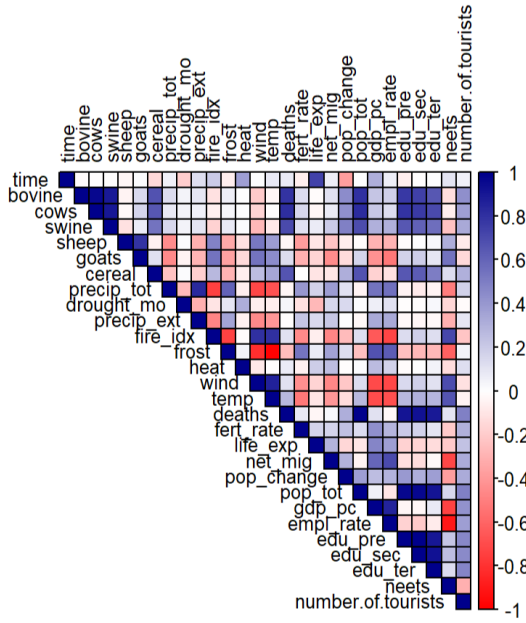
All data were downloaded from the respective websites and merged by NUTS code and year so that for each NUTS2 code we have the values of all our variables from 2000 to 2023.

Missing values were treated using the Last Observation Carried Forward (LOCF) method. This means that when a missing value was found in the dataset, it was replaced with the last available value for that variable, i.e. the value for the previous year and the same NUTS2 code.

It was decided to treat missing values using this technique as there were relatively few missing observations (only 6.68% of the total) and given that the typical temporal correlation of time series was present, making it sensible to use the previous value to estimate a missing observation.

After checking for multicollinearity, some variables were removed or modified. Due to high correlations among several variables, some variables were removed or modified. Particular importance was given to correlations that exceeded the threshold of 0.7 in absolute terms. Taking this into consideration, the following adjustments have been made:

- The variables concerning educational level were combined to create only two levels: "up to upper secondary education" and "higher than secondary education".
- All population-related variables (e.g., deaths, education, etc.) were divided by



Correlation matrix

Var1	Var2	Correlation
edu_sec	edu_pre	0.9904200
temp	frost	-0.9831891
pop_tot	deaths	0.9809224
edu_pre	pop_tot	0.9672232
edu_sec	pop_tot	0.9657062
cows	bovine	0.9623537
edu_sec	deaths	0.9266882
edu_ter	edu_sec	0.9249035
edu_pre	deaths	0.9168890
edu_ter	pop_tot	0.9148645
swine	cows	0.9086589
edu_ter	edu_pre	0.9084703
neets	empl_rate	-0.9072271
edu_ter	deaths	0.8992679
swine	bovine	0.8955805
temp	wind	0.8694148
empl_rate	gdp_pc	0.8539814
precip_ext	precip_tot	0.8351325
pop_tot	cows	0.8229020
wind	frost	-0.8153526
deaths	cows	0.8133866
pop_tot	bovine	0.8080740
wind	fire_idx	0.8062322
temp	fire_idx	0.8005908
deaths	bovine	0.7985208
goats	sheep	0.7935065
edu_pre	cows	0.7907619
edu_pre	bovine	0.7746313
edu_sec	cows	0.7676977
edu_sec	bovine	0.7458130
fire_idx	precip_tot	-0.7452092
neets	gdp_pc	-0.7383757
frost	fire_idx	-0.7382521
life_exp	time	0.7327575
empl_rate	fire_idx	-0.7215838
neets	net_mig	-0.7189938
empl_rate	wind	-0.7142298
wind	precip_tot	-0.7134618
neets	fire_idx	0.7133537
empl_rate	net_mig	0.7040753
deaths	swine	0.7018020
edu_ter	cows	0.7013770
gdp_pc	wind	-0.7000400

Pairs of variables with high correlation

the total population to obtain rates.

- Total population change has been removed as it is already represented by other included variables.
- The variables "goat" and "sheep" have been combined into a single variable.
- The variable "NEETs" (Not in Education, Employment, or Training) has been removed.
- The variable "total precipitation" has been removed, giving priority to the variable "extreme total precipitation" as it is considered a more relevant indicator.

These operations led to the final dataset, used to perform the analysis. A description of the variables included in the latter is provided in Table 1.

Variable	Description
time	Year of observation
nuts	Nuts2 code for the district
region	Name of the Italian region
zone	Macro-geographical zone (north, center, south)
GDP.pro.capita	Gross Domestic Product per capita
bovine	Number of bovine animals (in thousands)
ovine	Number of ovine animals (in thousands)
swine	Number of swine animals (in thousands)
cereals_grain	Cereal grain production (1000 ha)
months.of.droughts	Number of drought months in the year
total.extreme.precipitation.mm	Total extreme precipitation in mm
fire.weather.index	Index representing fire-prone weather conditions
frost.days	Number of frost days in the year
heatdays	Number of days classified as heatwaves in the year
mean.wind.speed	Average wind speed (m/s)
temperature	Annual mean temperature
fertility.rate	Number of children per woman
life.expectancy	Life expectancy at birth (in years)
Crude.rate.of.net.migration	Net migration rate (per 1000 people)
employment.rate	Employment rate (percentage of working-age population)
death.rate	Crude death rate (per 1000 people)
number.of.tourists.rate	Number of tourists (per inhabitant)
low_and_medium_education.rate	Share with low or medium education (per inhabitant)
high_education.rate	Share with high education (per inhabitant)

Table 1: Description of the variables in the final dataset

Analysis: primary research question

Methodology

To answer the primary research question it was necessary to create a model that had "GDP per capita" as the response and the other factors as predictors, in order to verify whether the abovementioned variables had an impact on the economy of each region. Since the data are structured hierarchically, with observations nested within regions over time, using a random effects model is a sensible choice in this analysis. Economic indicators such as GDP per capita are in fact influenced not only by time and other predictive factors, but also by unobservable regional characteristics that can cause variations in baseline GDP levels and growth trends. By allowing both the intercept and the effect of time to differ across regions or zones, the model accounts for this heterogeneity, improves model fit, and leads to more accurate and generalisable inferences about how economic and environmental factors influence GDP in different parts of Italy.

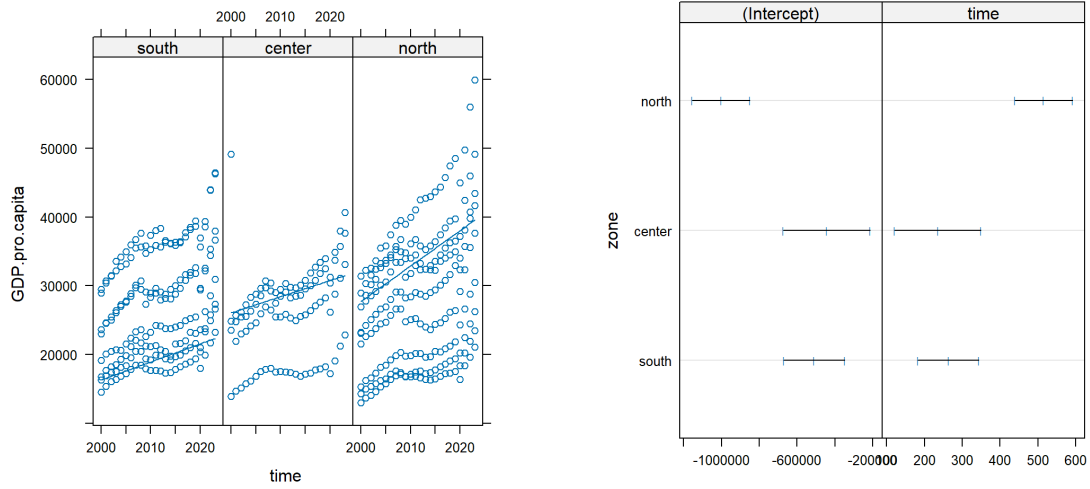
In this context, normality is assumed when modelling GDP per capita, even though it is known that this is not fully satisfied. To allow the use of other distributions in the modelling, the class of generalised mixed models could be used. This generalisation is not treated in this project, as the advantages of a more correct specification of the model are not sufficient to justify a much more complex framework to manage, which would cause difficulties for hypothesis testing and raise many other challenges.

Model specification

The regions were grouped into three zones: north, centre and south. This grouping was used to define the model, which included a random intercept for each zone and a random slope between time and each zone.

By fitting a linear model for each zone, it is possible to estimate the random factors and understand whether it is necessary to include them in the model or not (see Figure 4a). It was clear that both a random intercept and a random slope had to be included because, by grouping the data by zone, it was possible to obtain the graph presented in Figure 4b, which shows a confidence interval for each random effect. Since these random effects for the intercepts and slopes are not completely overlapping, it is worth to include these factors in the model. The model is then fitted including all variables as predictors. Those variables that are not statistically significant are still kept in the model as confounders. Among the climatic variables, the only one that appears to be significant, with a p-value of 0.014, is "total extreme precipitation".

The summary of the model fitted is presented in Figure 5.



(a) Linear models per zone

(b) Confidence Intervals for random effects

Figure 4

```

Linear mixed-effects model fit by maximum likelihood
Data: data_model_zone
      AIC      BIC    logLik
9388.664 9494.228 -4669.332

Random effects:
Formula: ~time | zone
Structure: General positive-definite, Log-Cholesky parametrization
              StdDev      Corr
(Intercept) 1.288349e-02 (Intr)
time         8.506377e-01 0.845
Residual    2.521976e+03

Fixed effects: GDP.pro.capita ~ time + bovine + ovine + swine + cereals_gra
in + months.of.droughts + total.extreme.precipitation.mm + fire.weathe
r.index + frost.days + heatdays + mean.wind.speed + temperature + ferti
lity.rate + life.expectancy + Crude.rate.of.net.migration + employment.
rate + death.rate + number.of.tourists.rate + low_and_medium_education.
rate + high_education.rate
              Value Std.Error DF   t-value p-value
(Intercept) -1143540.0 78309.98 481 -14.602736 0.0000
time          625.9    47.15 481  13.276269 0.0000
bovine        -1.0     1.14 481  -0.878489 0.3801
ovine         -0.6     0.29 481  -1.999596 0.0461
swine          1.4     0.35 481   3.996597 0.0001
cereals_grain -7.5     2.00 481  -3.751089 0.0002
months.of.droughts 35.9  80.77 481   0.444210 0.6571
total.extreme.precipitation.mm -3.3  1.35 481  -2.466401 0.0140
fire.weather.index -27.0 50.55 481  -0.534905 0.5930
frost.days     18.6   18.94 481   0.982003 0.3266
heatdays       9.2   51.41 481   0.178841 0.8581
mean.wind.speed 310.9 525.95 481   0.591171 0.5547
temperature    271.8  310.66 481   0.875030 0.3820
fertility.rate 2895.9 1346.09 481   2.151320 0.0319
life.expectancy -1371.0 218.46 481  -6.275901 0.0000
Crude.rate.of.net.migration  48.2  45.87 481   1.051756 0.2934
employment.rate 557.5  45.69 481  12.201203 0.0000
death.rate    -1358679.9 167327.75 481  -8.119872 0.0000
number.of.tourists.rate  48.1  17.13 481   2.809401 0.0052
low_and_medium_education.rate -6125.7 3240.44 481  -1.890405 0.0593
high_education.rate  945.8  8976.37 481   0.105361 0.9161

```

Figure 5: Summary of the random model for the 3 zones (north, centre, south)

From the model it is possible to extract the coefficients for the random intercept and the random slope for each zone (see Table 2).

Zone	Intercept	Time
North	0.01149	0.89842
Center	0.00297	0.23223
South	-0.01446	-1.13064

Table 2: Random intercepts and slopes by zone

It can be seen that the south is disadvantaged: its intercept is in fact negative, while those of the centre and north are positive. The intercept of the north is the one with the highest value.

However, the most significant part concerns the random slope. Again, this value turns out to be negative for the south. This means that although the south was already more disadvantaged, this gap has increased over time: GDP per capita has decreased in these areas. On the contrary, for the centre and north, the slope is positive, and once again the north shows the highest improvement over time.

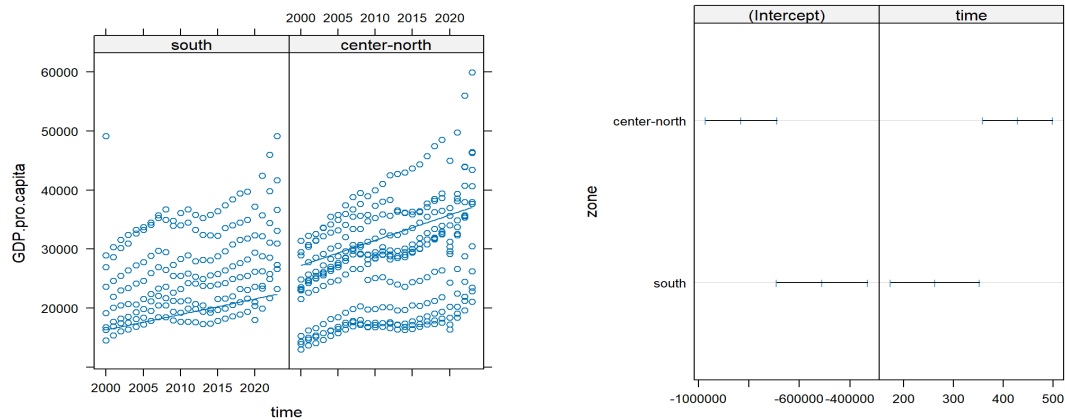
To understand whether these differences are significant, it is necessary to test whether the random slopes for the different zones are statistically different. In practical terms, it is sufficient to compare the model already implemented with an identical one that does not include a random slope (but only the random intercept).

It is being tested whether the variance of the random slope is different from zero. This hypothesis coincides with the boundary condition of the variance (since the variance must always be positive), and this does not allow us to apply Wilks' theorem. The convergence of the Likelihood Ratio Test (LRT) is therefore not guaranteed. To approximate the distribution of the LTR, a mixture of chi-squared distributions must be used.

The p-value for the test of interest in this case is 0.3057. This means that there is no evidence of a significant difference in the random slopes for the three different areas.

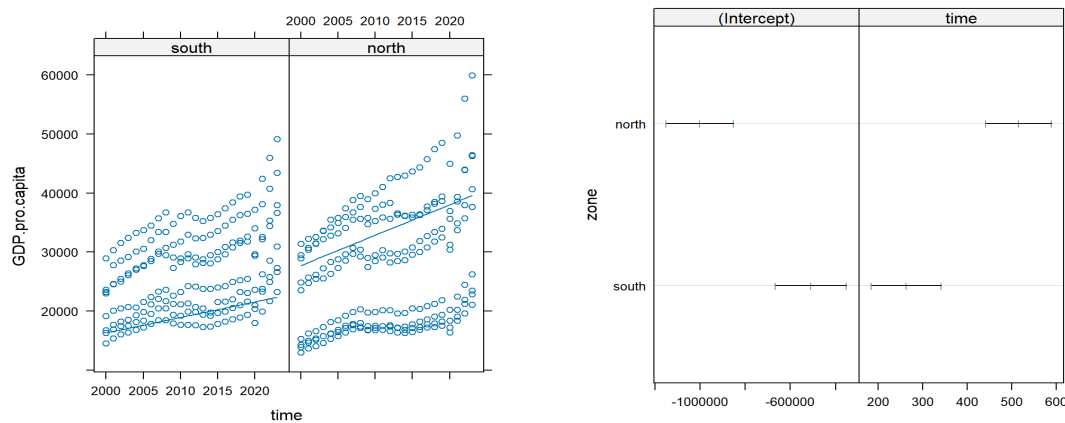
Model adjustments

In this section, the previous analysis is repeated in two different versions. In the first, the centre and north are combined into a single group (Model A). In the second, the north and south are compared, excluding the centre from the analysis (Model B).



Zone	Intercept	Time
Center-North	0.02368	0.69213
South	-0.02368	-0.69213

Model A



Zone	Intercept	Time
North	0.00139	0.40693
South	-0.00139	-0.40693

Model B

It can be seen that there are no huge differences between the initial model and these two variants. It should be pointed out, however, that if the Centre is excluded from the analysis, the difference between North and South becomes more significant.

This can be observed not only from the random coefficient values, but also from the p-values of the tests for the significance of the random slope in the different models.

In Model A the p-value is 0.2904, while for Model B it is 0.2714. These values are still far from being statistically significant, but they have reduced compared to those of the initial model.

Even in terms of the significance of climatic variables in the two variants, there are minor changes. In model A, as in the original model, the only significant climate variable is "total extreme precipitation" with a p-val of 0.0567.

In Model B, three climate variables turn out to be significant: "total extreme precipitation" (p-val=0.0077), "mean wind speed" (p-val=0.0003) and "temperature" (p-val=0.0265). Not only a higher number of climate variables result to be significant, but their p-values are also remarkably reduced.

Model by region

Given the results of previous analyses, it was decided to fit a model that included the random effects already discussed, but instead of at the zone level, at the regional level.

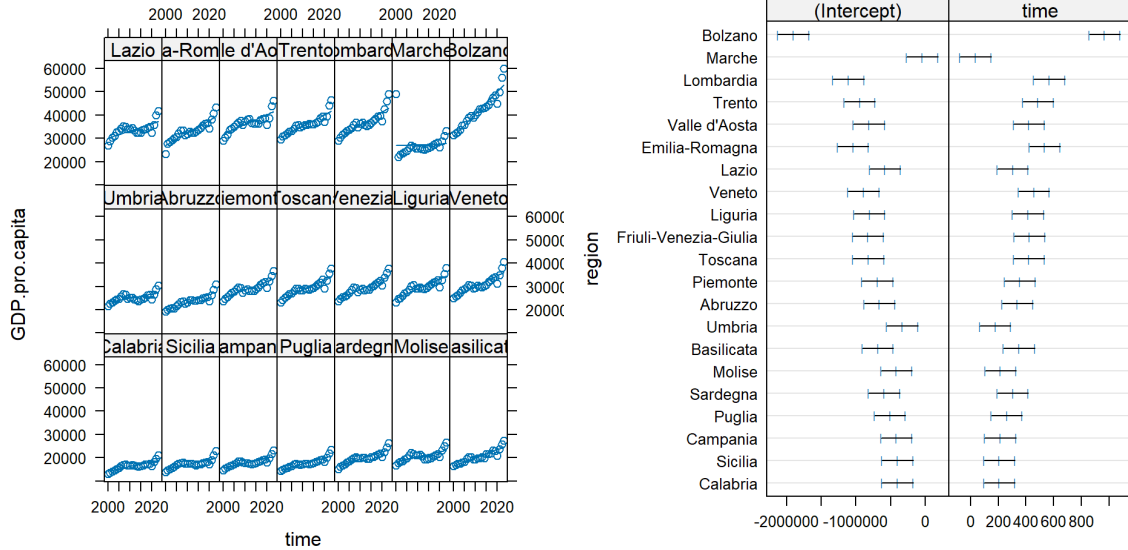


Figure 8: Model by Region

Table 3: Random Effects for each Regions

Region	Abruzzo	Basilicata	Bolzano	Calabria	Campania	Emilia-Romagna	Friuli-Venezia-Giulia	Lazio	Liguria
Intercept	-0.00767	-0.00098	0.00907	0.00116	0.00550	-0.00255	-0.00619	0.01604	0.00643
Time	-1.48782	-0.19018	1.75902	0.22475	1.06632	-0.49367	-1.20114	3.11045	1.24726

Region	Lombardia	Marche	Molise	Piemonte	Puglia	Sardegna	Sicilia	Toscana	Trento	Umbria	Valle d'Aosta	Veneto
Intercept	0.00482	-0.00797	-0.00199	-0.00426	0.00163	-0.00779	0.00654	-0.00364	0.00122	-0.01223	0.00486	-0.00199
Time	0.93522	-1.54524	-0.38698	-0.82617	0.31597	-1.51187	1.26948	-0.70689	0.23627	-2.37176	0.94284	-0.38585

In Figure 8 and in Table 3 is possible to see the results obtained with this new approach. One more time, the graph showing the confidence intervals for the random effects of each region emphasises the need to include the random effects in the model, also at regional level. It can be observed that, in the individual linear models fitted for each region, the random slopes and intercepts vary considerably from region to region. If it is of interest to have more details and to investigate differences at the regional level, and not only by dividing them into north, centre and south, it is possible to look at the estimates of the random coefficients and random slopes for each region (see Table 3). As expected, both tend to show negative values for southern regions and positive values for northern regions. However, it is interesting to note that each area is broken down into sub-parts: the regions. By construction of the model, adding up the estimates of all the random slopes of the southern regions results in the estimate of the random slope for the southern area of the original model.

	Value	Std.Error	DF	t-value	p-value
(Intercept)	-782903.4	98890.68	463	-7.916857	0.0000
time	392.0	60.41	463	6.488528	0.0000
bovine	-4.1	1.67	463	-2.430488	0.0155
ovine	1.1	0.78	463	1.391712	0.1647
swine	1.5	0.79	463	1.864365	0.0629
cereals_grain	5.0	2.94	463	1.698473	0.0901
months.of.droughts	43.1	59.90	463	0.719744	0.4720
total.extreme.precipitation.mm	-2.1	1.02	463	-2.059390	0.0400
fire.weather.index	-115.2	43.63	463	-2.640310	0.0086
frost.days	6.1	16.13	463	0.377577	0.7059
heatdays	25.1	38.76	463	0.646553	0.5182
mean.wind.speed	-2080.0	947.78	463	-2.194644	0.0287
temperature	189.8	289.74	463	0.655151	0.5127
fertility.rate	725.1	1252.20	463	0.579025	0.5629
life.expectancy	-50.6	267.09	463	-0.189270	0.8500
Crude.rate.of.net.migration	177.3	35.84	463	4.947526	0.0000
employment.rate	565.7	47.78	463	11.838419	0.0000
death.rate	-485658.3	244500.72	463	-1.986327	0.0476
number.of.tourists.rate	46.4	13.75	463	3.375764	0.0008
low_and_medium_education.rate	-4040.3	2477.97	463	-1.630474	0.1037
high_education.rate	-6033.1	7401.44	463	-0.815119	0.4154

Figure 9: Summary of the random model at regional level

As illustrated by the summary output of the model presented in Figure 9, if the analysis is carried out at regional level, several statistically significant climate variables emerge, namely "total extreme precipitation" (p-value = 0.0400), "fire weather index" (p-value = 0.0086) and "mean wind speed" (p-value = 0.0287).

This means that, as already pointed out in the extensive literature on the topic, climate change does indeed have an impact on GDP per capita, which changes differently over time between northern and southern Italy and, more specifically, among the different regions.

In particular, if we look at the coefficients of the climate variables that are significantly different from zero, these values are all negative. The impact of climate change is thus negative on economic well-being.

More specifically, a unitary increase in "total extreme precipitation" is associated with a decrease in "GDP per capita" by 2.1 units, on average, holding other variables constant. This negative effect may reflect economic disruption due to extreme weather events such as floods or storms.

An increase of one unit in the "fire weather index" leads to a decrease of about 115.2 units in "GDP per capita", on average. This suggests that areas with higher fire risk tend to have lower economic output, possibly due to the destructive impact of wildfires on infrastructure, agriculture, and productivity.

A unitary increase in "mean wind speed" is associated with a decrease in "GDP per capita" by 2,080 units, on average. This strong negative relationship reflects the damaging effects of strong winds on regional economies.

These findings highlight that extreme or hazardous weather conditions consistently lead to lower economic output due to immediate destruction, ongoing disruption, and the diversion of resources away from growth and development. Negative economic impacts are tangible, immediate and, in many cases, long-lasting, especially for the most vulnerable regions and populations, as also highlighted by several publications[7][16].

Analysis: secondary research question

This part of the analysis explores whether there have been noticeable changes over time in the average temperature in Italy, in connection with the Paris Agreement.

To do this, the average annual temperature was first calculated for different regions of Italy (north, centre, and south) as well as for the country as a whole. These values were then brought together and displayed (see Figure 10), making it easier to compare regional trends and observe how temperatures have evolved over the years. The focus then shifted to the national average, where a more detailed analysis was carried out to detect any significant shifts or turning points in the temperature trend. These potential shifts, known as breakpoints, were identified using a statistical method that highlights

where meaningful changes in the data may have occurred. The results were visualized (see Figure 11), showing both the observed temperatures and the points in time where the trend appears to change, helping to better understand how Italy’s climate may have responded to global and national actions.

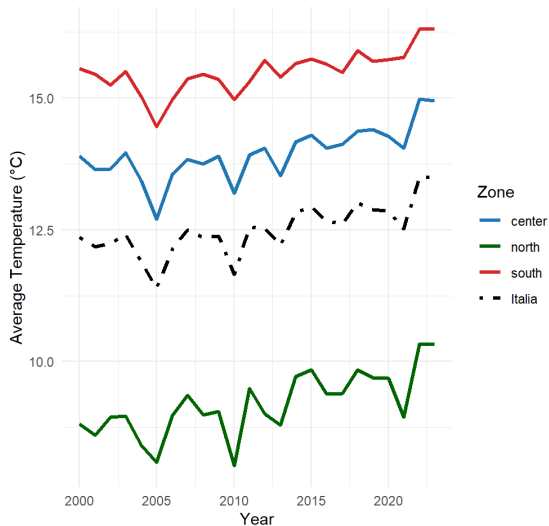


Figure 10: Average temperature by zone

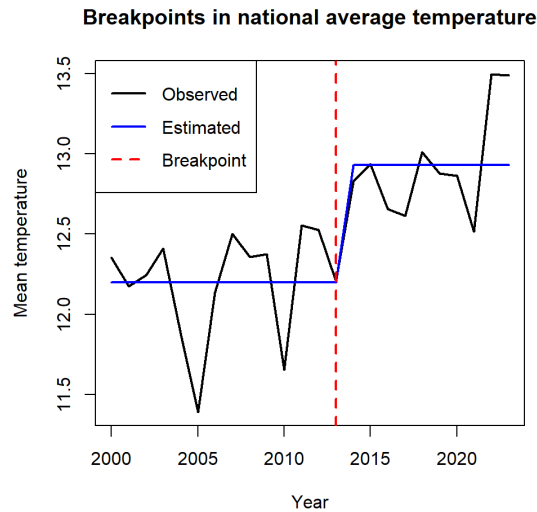


Figure 11: Breakpoints analysis

m = 1						2013	
m = 2						2013	2020
m = 3	2003	2006				2013	
m = 4	2003	2006				2013	2020
m = 5	2003	2006	2010			2013	2020
m = 6	2003	2006	2010	2013	2017		2020
Fit:							
m	0	1	2	3	4	5	6
RSS	5.620	2.504	2.261	1.909	1.666	1.593	1.550
BIC	39.625	26.573	30.478	32.784	35.873	41.152	46.849

Figure 12: Summary output of the breakpoints analysis

As shown in the summary output in Figure 12, the optimal number of breakpoints estimated in the historical series of annual temperatures in Italy was 1, as this presents the minimum value for the BIC (BIC=26.573). The estimated breakpoint is set in 2013, well before the Paris Agreement entered into force. The conclusion that can be drawn from this analysis is therefore that the Paris Agreement has not (yet) had a statistically significant impact on the average temperature trend in Italy. It is also interesting to note that if two breakpoints were considered (second best

option, with $BIC=30.478$), these two would be placed in 2013 and 2020. The estimated breakpoint for 2020 is probably due to the COVID pandemic, which forced everyone to stay at home, reducing human polluting activities and leading to a change in the temperature trend. However, it is more difficult to attribute the 2013 breakpoint to a well defined factor.

Conclusions

This project aims to verify whether climate change is contributing to the persistence or worsening of economic and social disparities between northern and southern Italy. Starting from a well documented structural gap in terms of GDP per capita, employment and infrastructure, the analysis focused on how climate variables interact with these existing differences. Using a linear mixed model that incorporates regional and zone random effects, it emerged that southern regions not only start from a less favourable economic base, but also show a tendency towards slower or even negative growth over time, particularly when climate variables are taken into account. Although the differences in the model's random slopes between zones were not statistically significant at conventional thresholds, southern Italy consistently showed negative trends in GDP per capita compared to central and northern regions.

This dynamic became even clearer in the regional model, where the estimated coefficients confirmed that southern regions are disproportionately affected by adverse climatic conditions. In particular, variables such as "total extreme precipitation", "wind speed" and "fire risk index" had statistically significant and negative impacts on GDP per capita, suggesting that climate shocks are indeed an additional burden on already disadvantaged areas.

The analysis also examined whether the implementation of the Paris Agreement had marked a turning point in temperature trends in Italy. Using a breakpoint analysis on the national temperature series, only one statistically significant change was identified, occurring in 2013, before the Paris Agreement came into force. No evidence was therefore found that the Agreement itself produced detectable changes in Italy's climate trends. This highlights both the long delay between policy implementation and observable effects, and the limitations of relying solely on voluntary international commitments to produce immediate results. A second breakpoint in 2020, less evident in the analysis, may reflect the short-term effects of COVID-19 lockdowns.

In conclusion, the results of this project support the idea that climate change acts as a multiplier of existing inequalities, as already widely documented in the existing literature. Southern Italy appears not only to be more exposed to climate-related risks, but also less equipped to mitigate or adapt to them, thus reinforcing its economic vulnerability. At the same time, the absence of measurable effects of the Paris Agreement

on national temperature trends calls for reflection on the scope and application of climate policies. If climate change continues to have a disproportionate impact on the South without more incisive interventions and adaptation strategies, the historical gap between North and South risks becoming even more entrenched in the coming decades.

Limitations and future work

Although the results of this project offer important insights into the interaction between climate change and regional economic disparities in Italy, it is necessary to acknowledge some limitations.

Firstly, the analysis is based on annual data, which may mitigate short-term shocks and delay the identification of immediate climate impacts.

Secondly, although the models include a wide range of socio-economic and climate variables, other relevant factors, such as infrastructure resilience, institutional quality or regional policy interventions, have not been directly incorporated due to data availability.

The use of the LOCF method for handling missing values, although justifiable given the data structure, may introduce some bias.

Furthermore, while the use of random effects models allowed for the inclusion of unobservable regional heterogeneity, the statistical power to detect differences in slopes between areas may have been limited by the sample size or the complexity of the model. Future work could address these limitations by exploring high-frequency or satellite-derived climate data, integrating additional indicators of adaptive capacity, and adopting more flexible models, such as non-linear models or machine learning approaches.

It would also be useful to study the role of specific policy measures adopted at the regional level in response to climate risks and to compare Italian regional dynamics with those of other European countries facing similar internal disparities.

Finally, with the availability of additional post-Paris Agreement data in the coming years, it will be essential to reassess whether international climate policies are beginning to produce measurable effects at the national or subnational level.

Appendix

In this section, all the code used for the analysis is provided, ensuring the reproducibility and transparency of the work carried out.

```
1  # Libraries ####
2
3  library(tidyverse)
4  library(nlme)
5  library(lattice)
6  library(glmnet)
7  library(corrplot)
8  library(ggplot2)
9  library(strucchange)
10 library(curl)
11 library(sf)
12 library(eurostat)
13 library(giscoR)
14 library(magick)
15
16 # Data Manipulation ####
17
18 # Read dataset
19
20 data<-read_csv("DATA.csv")
21 data$GDP.in.millions<-1000000*1000*data$GDP.in.millions/data$tot
   .population
22 data <- data %>% rename(GDP.pro.capita = GDP.in.millions)
23
24 # Dealing with Employment rate
25 data$employment.rate <- sapply(data$employment.rate, function(x)
   {
26   x <- as.character(x)
27   x <- gsub(",", "", x)
28   n <- nchar(x)
29   if (n == 3) {
30     paste0(substr(x, 1, 2), ".", substr(x, 3, 3))
31   } else if (n == 2) {
32     paste0(x, ".0")
33   } else {
34     NA
35   }
}
```

```

36 })
37
38 # Creation of variable zone (north, center, south)
39
40 center <- c("ITI1", "ITI2", "ITI3", "ITI4")
41 north <- c("ITC1", "ITC2", "ITC3", "ITC4", "ITH1", "ITH2", "ITH3",
42           "ITH4", "ITH5")
43 south <- c("ITF1", "ITF2", "ITF3", "ITF4", "ITF5", "ITF6", "ITG1",
44           "ITG2")
45
46 data$zone <- with(data, ifelse(nuts %in% center, "center",
47                               ifelse(nuts %in% south, "south",
48                                       ifelse(nuts %in% north, "north", NA))))
49 data <- data %>%
50   relocate(zone, .after = region)
51
52 # CHOOSE DATASET ####
53 # Uncomment and run only one of the following for choosing the
54 # dataset
55
56 ## No center
57
58 # data <- data[data$zone != "center", ]
59
60 ## Center and north joined
61
62 # center_north<-c(center,north)
63 # data$zone <- with(data, ifelse(nuts %in% center_north, "center-
64 #   north",
65 #                               ifelse(nuts %in% south, "south", NA)))
66
67 ## north, center, south
68
69 # data<-data
70
71 # zone as factor
72 data$zone <- as.factor(data$zone)
73 data$nuts <- as.factor(data$nuts)
74 data$region <- as.factor(data$region)
75 data$number.of.tourists <- as.numeric(gsub("\\.", "", data$

```

```

    number.of.tourists))
74 data$employment.rate<-as.numeric(data$employment.rate)
75 ## Dealing with NAs ####
76
77 sum(is.na(data))/14000 # 6% of observations are missing values
78
79 # Implementation of Last Observation Carried Forward (LOCF)
    method to impute missing values
80 # for variables
81
82 data <- data %>%
83   fill(everything(), .direction = "down")
84
85 sum(is.na(data))
86
87
88 ## Exploratory analysis and variables selection ####
89
90 ###Descriptive graphs ###
91
92 nuts2 <- get_eurostat_geospatial(output_class = "sf", resolution
    = "20", nuts_level = "2") %>%
93   filter(CNTR_CODE == "IT") %>%
94   rename(nuts = NUTS_ID)
95
96 data_gdp <- data[, c("time", "nuts", "region", "zone", "GDP.pro.
    capita")]
97
98 map_gdp <- nuts2 %>%
99   left_join(data_gdp, by = "nuts")
100
101 map_gdp$zone <- factor(map_gdp$zone, levels = c("north", "center
    ", "south"))
102
103 zone_coords <- data.frame(
104   zone = c("north", "center", "south"),
105   lon = c(9.5, 12.5, 15.5),
106   lat = c(45.5, 43, 41))
107
108 gdp_zone_time <- data %>%
109   group_by(zone, time) %>%
110   summarise(mean_gdp = round(mean(GDP.pro.capita, na.rm = TRUE)

```

```

110         ,0), .groups = "drop")
111
112 zone_coords_time <- zone_coords %>%
113   left_join(gdp_zone_time, by = "zone") %>%
114   filter(!is.na(mean_gdp))
115
116 temp_region <- data %>%
117   group_by(region, time) %>%
118   summarise(
119     mean_temp = mean(temperature, na.rm = TRUE),
120     zone = first(zone),
121     nuts = first(nuts),
122     .groups = "drop"
123   )
124
125 map_temp_year <- left_join(nuts2, temp_region, by = "nuts")
126
127 p1 <- ggplot(map_gdp) +
128   geom_sf(aes(fill = zone), color = "gray40", size = 0.2) +
129   scale_fill_manual(values = c(
130     "north" = "#84CCBD",
131     "center" = "#F9D340",
132     "south" = "#E86E78"
133   )) +
134   geom_text(
135     data = zone_coords_time,
136     aes(x = lon, y = lat, label = mean_gdp),
137     fontface = "bold", size = 4
138   ) +
139   labs(title = 'GDP per Capita by Zone: {closest_state}', fill =
140         "Zone") +
141   transition_states(time, transition_length = 1, state_length =
142     1) +
143   ease_aes('linear') +
144   theme_minimal() +
145   theme(
146     axis.text = element_blank(),
147     axis.ticks = element_blank(),
148     axis.title = element_blank(),
149     panel.grid = element_blank()
150   )

```

```

150 anim1 <- animate(p1, width = 800, height = 600, duration = 20,
    renderer = gifski_renderer())
151 anim_save("gdp_map.gif", anim1)
152
153 p2 <- ggplot(map_temp_year) +
154   geom_sf(aes(fill = mean_temp), color = "white") +
155   scale_fill_viridis_c(option = "plasma", name = "Avg Temp (C)")
    +
156   labs(title = 'Average Temperature: {closest_state}') +
157   transition_states(time, transition_length = 2, state_length =
    1) +
158   ease_aes('linear') +
159   theme_minimal() +
160   theme(
161     axis.text = element_blank(),
162     axis.ticks = element_blank(),
163     axis.title = element_blank(),
164     panel.grid = element_blank()
165   )
166
167 png_renderer <- file_renderer(
168   dir = "frames",
169   prefix = "gdp_frame_")
170 animate(p1, nframes = 100, fps = 10, renderer = png_renderer)
171
172 png_renderer <- file_renderer(
173   dir = "frames",
174   prefix = "temp_frame_")
175 animate(p2, nframes = 100, fps = 10, renderer = png_renderer)
176
177
178 ### Check for multicollinearity (only numeric variables) ###
179 num_vars <- data %>% select(where(is.numeric))
180
181 short_names <- c(
182   "time" = "time",
183   "bovine" = "bovine",
184   "cows" = "cows",
185   "swine" = "swine",
186   "sheep" = "sheep",
187   "goats" = "goats",
188   "cereals_grain" = "cereal",

```

```

189 "tot.precipitation" = "precip_tot",
190 "months.of.droughts" = "drought_mo",
191 "total.extreme.precipitation.mm" = "precip_ext",
192 "fire.weather.index" = "fire_idx",
193 "frost.days" = "frost",
194 "heatdays" = "heat",
195 "mean.wind.speed" = "wind",
196 "temperature" = "temp",
197 "deaths" = "deaths",
198 "fertility.rate" = "fert_rate",
199 "life.expectancy" = "life_exp",
200 "Crude.rate.of.net.migration" = "net_mig",
201 "Total.population.change" = "pop_change",
202 "tot.population" = "pop_tot",
203 "GDP.pro.capita" = "gdp_pc",
204 "employment.rate" = "empl_rate",
205 "Pre.primary.education" = "edu_pre",
206 "Primary..lower.and.upper.secondary.education" = "edu_sec",
207 "First.and.second.stage.of.tertiary.education" = "edu_ter",
208 "NEETS.." = "needs"
209 )
210
211 num_vars_short <- num_vars %>% rename(!!!setNames(names(short_
    names), short_names))
212
213 cor_matrix <- cor(num_vars_short, use = "complete.obs")
214
215 corrplot(cor_matrix,
216           method = "color",
217           tl.col = "black",
218           type = "upper",
219           tl.cex = 0.8,
220           addCoef.col = NULL,
221           col = colorRampPalette(c("red", "white", "darkblue"))
                (200),
222           mar = c(1,1,1,1),
223           addgrid.col = "black")
224
225
226 # Var with high corr
227 threshold <- 0.7
228 high_corr_pairs <- which(abs(cor_matrix) > threshold & lower.tri

```



```

229     (cor_matrix), arr.ind = TRUE)
230 results <- data.frame(Var1 = rownames(cor_matrix)[high_corr_
    pairs[,1]],
231   Var2 = colnames(cor_matrix)[high_corr_pairs[,2]],
232   Correlation = cor_matrix[high_corr_pairs])
233
234 var_high_corr <- results[order(abs(results$Correlation),
    decreasing = TRUE), ]
235 print(var_high_corr)
236
237 # Because of high correlation we:
238 # merge Pre primary education with variable up until Upper
    secondary education
239 # divide all population variables by the total population in
    that zone to get population rates:
240 # - deaths divide
241 # - remove total population change as it is already accounted
    for by rates and migration
242 # - create education variables and divide
243 # We choose ignore high correlation between temperature and
    other climate variables
244 # Bovine and cows, we remove cows
245 # Swine and cows we ignore the high correlation (maybe 1 animal
    variable?)
246 # We remove neets
247 # We remove total precipitation as we feel that total extreme is
    more important
248 # We merge goats and sheep
249
250 # Merge Pre-primary and Upper Secondary
251 data$low_and_medium_education <- data$Pre.primary.education +
252   data$Primary..lower.and.upper.secondary.education
253
254 # Divide deaths by total population
255 data$death.rate <- data$deaths / data$tot.population
256
257 # Remove total population change
258 data <- data %>% select(-Total.population.change)
259
260 # Divide education variables by total population
261 data$number.of.tourists.rate<-data$number.of.tourists / data$tot

```

```

    .population
262 data$low_and_medium_education.rate <- data$low_and_medium_
    education / data$tot.population
263 data$high_education.rate <- data$First.and.second.stage.of.
    tertiary.education / data$tot.population
264 data <- data %>% select(-cows)
265
266 data <- data %>% select(-NEETS..)
267 data <- data %>% select(-tot.precipitation)
268 data$ovine <- data$goats + data$sheep
269
270 # Clean up any variables no longer needed
271 data <- data %>%
272   select(-Pre.primary.education,
273     -Primary..lower.and.upper.secondary.education,
274     -low_and_medium_education,
275     -deaths,
276     -First.and.second.stage.of.tertiary.education,
277     -goats,
278     -sheep,
279     -tot.population,
280     -number.of.tourists)
281
282 data <- data %>%
283   relocate(ovine, .after = bovine)
284 data <- data %>%
285   relocate(GDP.pro.capita, .after = zone)
286
287 # Study the dependencies between GDP and each variable
288 numerical_vars <- names(data)[sapply(data, is.numeric)]
289
290 # Remove target variable
291 numerical_vars <- setdiff(numerical_vars, "GDP.pro.capita")
292
293 # P values for numerical variables
294 for (var in numerical_vars) {
295   formula <- as.formula(paste("GDP.pro.capita ~", var))
296   model <- lm(formula, data = data)
297   summary_model <- summary(model)
298
299   p_val <- summary_model$coefficients[2, 4]
300   coef <- summary_model$coefficients[2, 1]

```

```

301   r_squared <- summary_model$r.squared
302
303   cat(sprintf("%s: p-value = %.4f\n",
304             var, p_val))
305 }
306
307 write.csv(data, "Data_final.csv")
308
309 # Model ####
310
311 ## GDP ####
312
313 model <- lm(GDP.pro.capita ~.-nuts, data=data)
314
315 summary(model)
316
317 ## By region ####
318 data_model_region <- data[,-c(2,4)] #tolto nuts and zone (rimane
319 region)
320 data_mixed_region <- groupedData(GDP.pro.capita~time|region,
321   data=data_model_region)
322 plot(data_mixed_region)
323
324 lmList_region<-lmList(data_mixed_region)
325 plot(augPred(lmList_region),layout=c(7,3))
326 intervals(lmList_region)
327 plot(intervals(lmList_region))
328
329 data_mixed_region <- data_mixed_region %>%
330   mutate(across(where(is.numeric) & !any_of("time"), scale))
331
332 model_region_rand<-lme(fixed=GDP.pro.capita~time + bovine +
333   ovine + swine +
334   cereals_grain + months.of.droughts +
335   total.extreme.precipitation.mm +
336   fire.weather.index + frost.days +
337   heatdays + mean.wind.speed +
338   temperature + fertility.rate + life.
339   expectancy +
340   Crude.rate.of.net.migration +
341   employment.rate + death.rate +
342   number.of.tourists.rate + low_and_

```

```

336         medium_education.rate +
        high_education.rate, random=~time|region
337         ,
        data=data_model_region, method = "ML
        ")
338 summary(model_region_rand)
339 ranef(model_region_rand)
340
341 ## By zone ####
342 data_model_zone <- data[,-c(2,3)] # tolto nuts and region (
        rimane zone)
343 data_mixed_zone <- groupedData(GDP.pro.capita~time|zone, data=
        data_model_zone)
344 plot(data_mixed_zone)
345
346 lmList_zone<-lmList(data_mixed_zone)
347 plot(augPred(lmList_zone))
348 intervals(lmList_zone)
349 plot(intervals(lmList_zone))
350
351 data_mixed_zone <- data_mixed_zone %>%
352   mutate(across(where(is.numeric) & !any_of("time"), scale))
353
354 model_zone_rand<-lme(fixed=GDP.pro.capita~time + bovine + ovine
        + swine +
355         cereals_grain + months.of.droughts +
        total.extreme.precipitation.mm +
356         fire.weather.index + frost.days +
        heatdays + mean.wind.speed +
357         temperature + fertility.rate + life.
        expectancy +
358         Crude.rate.of.net.migration + employment.
        rate + death.rate +
359         number.of.tourists.rate + low_and_medium_
        education.rate +
360         high_education.rate, random=~time|zone,
        data=data_model_zone, method = "ML")
361 summary(model_zone_rand)
362
363 ranef(model_zone_rand)
364
365 ### testing the random slope ####

```

```

366
367 mod0zone<-update(model_zone_rand,random=~1|zone)
368
369 loglikratio<-anova(mod0zone,model_zone_rand)$L.Ratio[2]
370 p.value_zone<-0.5*(1-pchisq(loglikratio,0)) + 0.5*(1-pchisq(
    loglikratio,1))
371 p.value_zone
372
373 # Testing for breakpoints in temperature - Paris Agreement ####
374
375 # Compute average temperature per year and zone
376 zone_temp <- data %>%
377   group_by(zone, time) %>%
378   summarise(mean_temp = mean(temperature, na.rm = TRUE), .groups
    = "drop")
379
380 # Compute national average temperature per year
381 italy_temp <- data %>%
382   group_by(time) %>%
383   summarise(mean_temp = mean(temperature, na.rm = TRUE)) %>%
384   mutate(zone = "Italia")
385
386 # Combine both datasets
387 temp_plot_data <- bind_rows(zone_temp, italy_temp)
388 temp_plot_data$zone <- factor(temp_plot_data$zone,
    levels = c("center", "north", "
389               south", "Italia"))
390
391 # Plot
392 ggplot(temp_plot_data, aes(x = as.numeric(time),
393                             y = mean_temp,
394                             color = zone,
395                             linetype = zone)) +
396   geom_line(linewidth = 1.2) +
397   scale_color_manual(values = c("center" = "#1f77b4",
398                                 "north" = "darkgreen",
399                                 "south" = "#d62728",
400                                 "Italia" = "black")) +
401   scale_linetype_manual(values = c("center" = "solid",
402                                    "north" = "solid",
403                                    "south" = "solid",
404                                    "Italia" = "dotted")) +

```

```

405 labs(title = "",
406       x = "Year",
407       y = "Average Temperature (C)",
408       color = "Zone",
409       linetype = "Zone") +
410 theme_minimal()
411
412 # Extract only the Italy time series
413 italy_ts <- temp_plot_data %>%
414   filter(zone == "Italia") %>%
415   arrange(time)
416
417 # Convert to time series object
418 temp_ts <- ts(italy_ts$mean_temp, start = min(italy_ts$time),
419              frequency = 1)
420
421 # Breakpoint analysis
422 bp_model <- breakpoints(temp_ts ~ 1)
423
424 # Summary of breakpoints
425 summary(bp_model)
426
427 # Plot with breakpoints corresponding to the minimum BIC
428 plot(temp_ts, type = "l", col = "black", lwd = 2,
429       ylab = "Mean temperature", xlab = "Year",
430       main = "Breakpoints in national average temperature")
431 lines(fitted(bp_model), col = "blue", lwd = 2)
432
433 # Extract optimal number of breakpoints according to BIC
434 optimal_breaks <- breakpoints(temp_ts ~ 1, breaks = which.min(
435   BIC(bp_model)))
436
437 # Add vertical lines at BIC-optimal breakpoints
438 abline(v = time(temp_ts)[optimal_breaks$breakpoints], col = "red",
439        lwd = 2, lty = 2)
440
441 # Add legend
442 legend("topleft", legend = c("Observed", "Estimated", "
    Breakpoint"),
443       col = c("black", "blue", "red"), lwd = 2, lty = c(1, 1,
444       2))

```

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